Engineering Mathematics (AHT-003) - Unit 1

Unit 1 - Calculus I - notes in easy English, matched to the UTU syllabus.

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What this unit covers (syllabus): limit, continuity and differentiability; Rolle's and mean-value theorems; Maclaurin/Taylor expansions (one and two variables); partial differentiation; maxima-minima and Lagrange multipliers.

Key ideas: A function is **continuous** at a point when its limit equals its value there; **differentiable** means the derivative exists (differentiable → continuous, not the reverse). **Rolle's theorem:** if f is continuous on [a,b], differentiable on (a,b), and f(a) = f(b), then some c has f'(c) = 0. **Mean value theorem (Lagrange):** f(b) - f(a) = f'(c)(b - a). **Taylor/Maclaurin** expand a function as an infinite polynomial - memorise the standard series: e^x, sin x, cos x, log(1+x), $(1+x)^k$. **Partial derivatives** differentiate with respect to one variable, treating the rest as constants. For **maxima-minima** in two variables: solve fx = 0, fy = 0, then check rt - s^2 (positive with r < 0 gives a maximum, positive with r > 0 a minimum). **Lagrange multipliers** handle constrained problems: solve $\nabla f = \lambda \nabla g$.

Exam pointers: verify Rolle/LMVT for a given function (check the three conditions first - that is where marks hide), one Maclaurin expansion, and one maxima-minima of f(x,y) using the rt - s^2 test. Show every condition check explicitly.

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